

SAFETY DATA SHEET

according to Regulation (EC) No. 1907/2006



ROVIMIX® D3-500

5016016

Version 1.0

Revision Date 22.01.2016

Date of last issue: -

SECTION 1: Identification of the substance/mixture and of the company/undertaking

1.1 Product identifier

Trade name : ROVIMIX® D3-500

1.2 Relevant identified uses of the substance or mixture and uses advised against

Use of the Sub-stance/Mixture : Additive for animal nutrition in premixes and compound feeds

1.3 Details of the supplier of the safety data sheet

Company : DSM Nutritional Products Ltd.
PO Box 2676
4002 Basel

Telephone : +41618158888
Telefax : +41618157253
E-mail address of person responsible for the SDS : sds.nutritionalproducts@dsm.com

1.4 Emergency telephone number

+41 62 866 2314

SECTION 2: Hazards identification

2.1 Classification of the substance or mixture

Classification (REGULATION (EC) No 1272/2008)

Acute toxicity, Category 4	H302: Harmful if swallowed.
Specific target organ toxicity - repeated exposure, Category 2	H373: May cause damage to organs through prolonged or repeated exposure.
Chronic aquatic toxicity, Category 2	H411: Toxic to aquatic life with long lasting effects.

2.2 Label elements

Labelling (REGULATION (EC) No 1272/2008)

Hazard pictograms :



Signal word : Warning

Hazard statements	H302	Harmful if swallowed.
	H373	May cause damage to organs through prolonged or repeated exposure.
	H411	Toxic to aquatic life with long lasting effects.

Precautionary statements	Prevention:	
	P260	Do not breathe dust/ fume/ gas/ mist/ vapours/ spray.
	P264	Wash skin thoroughly after handling.
	P273	Avoid release to the environment.
	Response:	
	P314	Get medical advice/ attention if you feel unwell.
	P391	Collect spillage.
Disposal:		
P501	Dispose of contents/ container to an ap-	

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proved waste disposal plant.

Hazardous components which must be listed on the label:

67-97-0 cholecalciferol

Other hazards

Risk of dust explosion.

SECTION 3: Composition/information on ingredients

Brief description of the product : Mixture (preparation) containing active ingredient and auxiliary substances

3.1 Substances

Not applicable

3.2 Mixtures

Hazardous components

Chemical Name	CAS-No. EC-No. Registration number	Classification	Concentration (% w/w)
2,6-di-tert-butyl-p-cresol	128-37-0 204-881-4	Aquatic Acute 1; H400 Aquatic Chronic 1; H410	$\geq 2.5 - < 5$
cholecalciferol	67-97-0 200-673-2	Acute Tox. 2; H300 Acute Tox. 2; H330 Acute Tox. 3; H311 STOT RE 1; H372 Aquatic Chronic 4; H413	$\geq 1 - < 2.5$

For explanation of abbreviations see section 16.

SECTION 4: First aid measures

4.1 Description of first aid measures

- General advice : Move out of dangerous area.
Show this safety data sheet to the doctor in attendance.
- If inhaled : Move to fresh air.
Consult a physician after significant exposure.
- In case of skin contact : Take off contaminated clothing and shoes immediately.
Wash off with soap and plenty of water.
If symptoms persist, call a physician.
- In case of eye contact : Flush eyes with water as a precaution.
Remove contact lenses.
Protect unharmed eye.
Keep eye wide open while rinsing.
If eye irritation persists, consult a specialist.
- If swallowed : Clean mouth with water and drink afterwards plenty of water.
Do not give milk or alcoholic beverages.
Never give anything by mouth to an unconscious person.
Obtain medical attention.

4.2 Most important symptoms and effects, both acute and delayed

Symptoms : Acute overdose produces the following symptoms:

Nausea, Vomiting, Headache, Weakness, Abdominal pain,
Dry mouth, Metallic taste, Loss of appetite

4.3 Indication of any immediate medical attention and special treatment needed

Treatment : Treat symptomatically.

SECTION 5: Firefighting measures

5.1 Extinguishing media

Suitable extinguishing media : Water
Foam

5.2 Special hazards arising from the substance or mixture

Specific hazards during fire-fighting : Do not allow run-off from fire fighting to enter drains or water courses.

5.3 Advice for firefighters

Special protective equipment for firefighters : In the event of fire, wear self-contained breathing apparatus.

Further information : Collect contaminated fire extinguishing water separately. This must not be discharged into drains.
Fire residues and contaminated fire extinguishing water must be disposed of in accordance with local regulations.
Consider dust explosion hazard.

SECTION 6: Accidental release measures

6.1 Personal precautions, protective equipment and emergency procedures

Evacuate personnel to safe areas.
Use personal protective equipment.
Ensure adequate ventilation.
Avoid dust formation.
Avoid breathing dust.

6.2 Environmental precautions

Do not flush into surface water or sanitary sewer system.
Prevent further leakage or spillage if safe to do so.
If the product contaminates rivers and lakes or drains inform respective authorities.

6.3 Methods and material for containment and cleaning up

Pick up and arrange disposal without creating dust.
Keep in suitable, closed containers for disposal.

6.4 Reference to other sections

For personal protection see section 8.
For disposal considerations see section 13.

SECTION 7: Handling and storage

7.1 Precautions for safe handling

Advice on safe handling : Avoid formation of respirable particles.
Avoid exceeding the given occupational exposure limits (see

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section 8).
Avoid contact with skin and eyes.
For personal protection see section 8.
Dispose of rinse water in accordance with local and national regulations.
Do not breathe vapours/dust.
Smoking, eating and drinking should be prohibited in the application area.

Advice on protection against fire and explosion : Avoid dust formation. Provide appropriate exhaust ventilation at places where dust is formed. Take precautionary measures against static discharges.

Hygiene measures : When using do not eat or drink. When using do not smoke. Avoid contact with skin, eyes and clothing.

7.2 Conditions for safe storage, including any incompatibilities

Requirements for storage areas and containers : Protect against light.
Protect from humidity.

Keep container tightly closed and dry.

Recommended storage temperature : < 15 °C

7.3 Specific end use(s)

Specific use(s) : Not applicable

SECTION 8: Exposure controls/personal protection

8.1 Control parameters

Occupational Exposure Limits

Components	CAS-No.	Value type (Form of exposure)	Control parameters	Basis
2,6-di-tert-butyl-p-cresol	128-37-0	TWA	10 mg/m ³	GB EH40
Where no specific short-term exposure limit is listed, a figure three times the long-term exposure should be used				
cholecalciferol	67-97-0	TWA	0.005 mg/m ³	DSM Internal Limit

8.2 Exposure controls

Personal protective equipment

Eye protection : Safety glasses with side-shields

Hand protection : Consider the hazard characteristics of this product and any special workplace conditions when selecting the appropriate type of protective gloves. Glove material: for example nitrile rubber

Skin and body protection : Choose body protection according to the amount and concentration of the dangerous substance at the work place.

Respiratory protection : In the case of dust or aerosol formation use respirator with an approved filter.

SECTION 9: Physical and chemical properties

9.1 Information on basic physical and chemical properties

Appearance	: fine powder
Colour	: beige - light brown
Odour	: characteristic
Odour Threshold	: No information available.
pH	: No data available
Melting point/range	: not determined
Boiling point/boiling range	: not determined
Flash point	: Not applicable
Flammability (solid, gas)	: May form combustible dust concentrations in air
Vapour pressure	: Not applicable
Relative vapour density	: Not applicable
Density	: not determined
Water solubility	: dispersible
Partition coefficient: n-octanol/water	: Not applicable
Auto-ignition temperature	: No data available
Thermal decomposition	: Decomposes on heating. Potential for exothermic hazard
Explosive properties	: Not explosive
Oxidizing properties	: No data available

9.2 Other information

Combustibility index for deposited dust	: 3 (23 °C) : 3 (100 °C)
Dust explosion class	: St(H)1 (Product sample, Median value of the tested sample 0.06 mm, Loss on drying 2.3 %; The value was determined in the modified Hartmann tube.)
Minimum ignition energy	: 30 - 100 mJ (Product sample, Median value of the tested sample 0.06 mm, Loss on drying 2.3 %, EN 13821) The Minimum ignition energy (MIE) of a dust/air mix depends on the particle size the water content and the temperature of the dust. The finer and the dryer the dust the lower the MIE. : General remark: The indicated dust explosion characteristics are only valid for this product and are sensitive to the sample's parameters.

SECTION 10: Stability and reactivity

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10.1 Reactivity

No hazards to be specially mentioned.

10.2 Chemical stability

Stable under recommended storage conditions.

10.3 Possibility of hazardous reactions

Dust may form explosive mixture in air.

10.4 Conditions to avoid

Heat

10.5 Incompatible materials

Strong acids and strong bases

Strong oxidizing agents

10.6 Hazardous decomposition products

No decomposition if used as directed.

SECTION 11: Toxicological information

11.1 Information on toxicological effects

Acute oral toxicity	: Acute toxicity estimate : ca. 400 mg/kg (Calculation method)
Acute inhalation toxicity	: Acute toxicity estimate : 8 mg/l (Calculation method)
Acute dermal toxicity	: Acute toxicity estimate : > 5,000 mg/kg (Calculation method)
Skin irritation	: Prolonged skin contact may cause skin irritation.
Eye irritation	: Dust contact with the eyes can lead to mechanical irritation.
Sensitisation	: No known indication for sensitizing effect.
Carcinogenicity	: No indication for carcinogenicity known.
Genotoxicity in vivo cholecalciferol	: not genotoxic (Rat, Bone marrow, Mutagenicity (micronucleus test))
Teratogenicity cholecalciferol	: May lead to malformations at dose levels that cause maternal toxicity. NOAEL: 0.0095 mg/kg bw/d (Rabbit female, OECD Test Guideline 414)
STOT - repeated exposure cholecalciferol	: NOAEL (Oral, Rat) : 0.06 mg/kg bw/d

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Sub-chronic toxicity study (90-day)
(OECD Test Guideline 408)

Causes damage to organs through prolonged or repeated exposure.

Experience with human exposure
cholecalciferol

: RDA (Recommended Daily Allowance) 0.005 - 0.01 mg

Experience with human exposure: Ingestion

cholecalciferol

: Acute overdose produces the following symptoms:
Nausea, Vomiting, Headache, Weakness, Abdominal pain,
Dry mouth, Metallic taste, Loss of appetite

SECTION 12: Ecological information

12.1 Toxicity

Toxicity to daphnia and other aquatic invertebrates

2,6-di-tert-butyl-p-cresol : Daphnia magna (Water flea)
EC50 (48 h) 0.48 mg/l
(OECD Test Guideline 202)

No data is available on the product itself.

12.2 Persistence and degradability

Biodegradability

2,6-di-tert-butyl-p-cresol : Not readily biodegradable.
4.5 % (28 d)
(OECD Test Guideline 301C)

cholecalciferol

: Not readily biodegradable.
<= 7 % (28 d)
(OECD Test Guideline 301C)

No data is available on the product itself.

12.3 Bioaccumulative potential

Bioaccumulation

: No data available

Partition coefficient: n-
octanol/water

: Not applicable

12.4 Mobility in soil

Distribution among environ-
mental compartments

: No data available

12.5 Results of PBT and vPvB assessment

Assessment

: not determined

12.6 Other adverse effects

Additional ecological informa-
tion

: Toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment.

SECTION 13: Disposal considerations

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13.1 Waste treatment methods

- Product : Discharge into the environment must be avoided.
Do not contaminate ponds, waterways or ditches with chemical or used container.
Do not dispose of waste into sewer.
Offer surplus and non-recyclable solutions to a licensed disposal company.
- Contaminated packaging : Dispose of as unused product.
Do not re-use empty containers.

SECTION 14: Transport information

14.1 UN number

- ADN : UN 3077
- ADR : UN 3077
- RID : UN 3077
- IMDG : UN 3077
- IATA : UN 3077

14.2 UN proper shipping name

- ADN : ENVIRONMENTALLY HAZARDOUS SUBSTANCE, SOLID, N.O.S.
(2,6-di-tert-butyl-p-cresol, cholecalciferol)
- ADR : ENVIRONMENTALLY HAZARDOUS SUBSTANCE, SOLID, N.O.S.
(2,6-di-tert-butyl-p-cresol, cholecalciferol)
- RID : ENVIRONMENTALLY HAZARDOUS SUBSTANCE, SOLID, N.O.S.
(2,6-di-tert-butyl-p-cresol, cholecalciferol)
- IMDG : ENVIRONMENTALLY HAZARDOUS SUBSTANCE, SOLID, N.O.S.
(2,6-di-tert-butyl-p-cresol, cholecalciferol)
- IATA : Environmentally hazardous substance, solid, n.o.s.
(2,6-di-tert-butyl-p-cresol, cholecalciferol)

14.3 Transport hazard class(es)

- ADN : 9
- ADR : 9
- RID : 9
- IMDG : 9
- IATA : 9

14.4 Packing group

- ADN
- Packing group : III
- Classification Code : M7
- Hazard Identification Number : 90
- Labels : 9
- ADR

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Packing group : III
Classification Code : M7
Hazard Identification Number : 90
Labels : 9

RID

Packing group : III
Classification Code : M7
Hazard Identification Number : 90
Labels : 9

IMDG

Packing group : III
Labels : 9
EmS Code : F-A, S-F

IATA

Packing instruction (cargo aircraft) : 956
Packing instruction (passenger aircraft) : 956
Packing instruction (LQ) : Y956
Packing group : III
Labels : Miscellaneous Dangerous Goods

14.5 Environmental hazards

ADN

Environmentally hazardous : yes

ADR

Environmentally hazardous : yes

RID

Environmentally hazardous : yes

IMDG

Marine pollutant : yes

14.6 Special precautions for user

Not applicable

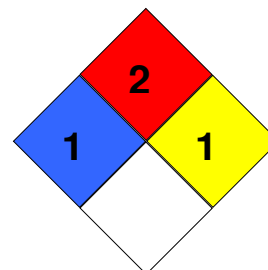
14.7 Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code

Remarks : Not applicable

SECTION 15: Regulatory information

15.1 Safety, health and environmental regulations/legislation specific for the substance or mixture

NFPA Classification : Health hazard: 1
Fire Hazard: 2
Reactivity Hazard: 1



15.2 Chemical Safety Assessment

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Not applicable

SECTION 16: Other information

Full text of H-Statements

- H300 : Fatal if swallowed.
- H311 : Toxic in contact with skin.
- H330 : Fatal if inhaled.
- H372 : Causes damage to organs through prolonged or repeated exposure.
- H400 : Very toxic to aquatic life.
- H410 : Very toxic to aquatic life with long lasting effects.
- H413 : May cause long lasting harmful effects to aquatic life.

Full text of other abbreviations

- Acute Tox. : Acute toxicity
- Aquatic Acute : Acute aquatic toxicity
- Aquatic Chronic : Chronic aquatic toxicity
- STOT RE : Specific target organ toxicity - repeated exposure

(Q)SAR - (Quantitative) Structure Activity Relationship; ADN - European Agreement concerning the International Carriage of Dangerous Goods by Inland Waterways; ADR - European Agreement concerning the International Carriage of Dangerous Goods by Road; ASTM - American Society for the Testing of Materials; bw - Body weight; CLP - Classification Labelling Packaging Regulation; Regulation (EC) No 1272/2008; DIN - Standard of the German Institute for Standardisation; ECHA - European Chemicals Agency; EC-Number - European Community number; ECx - Concentration associated with x% response; ELx - Loading rate associated with x% response; EmS - Emergency Schedule; ErCx - Concentration associated with x% growth rate response; GHS - Globally Harmonized System; IARC - International Agency for Research on Cancer; IATA - International Air Transport Association; IBC - International Code for the Construction and Equipment of Ships carrying Dangerous Chemicals in Bulk; IC50 - Half maximal inhibitory concentration; ICAO - International Civil Aviation Organization; IMDG - International Maritime Dangerous Goods; IMO - International Maritime Organization; ISO - International Organisation for Standardization; LC50 - Lethal Concentration to 50 % of a test population; LD50 - Lethal Dose to 50% of a test population (Median Lethal Dose); MARPOL - International Convention for the Prevention of Pollution from Ships; n.o.s. - Not Otherwise Specified; NO(A)EC - No Observed (Adverse) Effect Concentration; NO(A)EL - No Observed (Adverse) Effect Level; NOELR - No Observable Effect Loading Rate; OECD - Organization for Economic Co-operation and Development; OPPTS - Office of Chemical Safety and Pollution Prevention; PBT - Persistent, Bioaccumulative and Toxic substance; REACH - Regulation (EC) No 1907/2006 of the European Parliament and of the Council concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals; RID - Regulations concerning the International Carriage of Dangerous Goods by Rail; SADT - Self-Accelerating Decomposition Temperature; SDS - Safety Data Sheet; TRGS - Technical Rule for Hazardous Substances; UN - United Nations; vPvB - Very Persistent and Very Bioaccumulative; DSL - Domestic Substances List (Canada); KECI - Korea Existing Chemicals Inventory; TSCA - Toxic Substances Control Act (United States); AICS - Australian Inventory of Chemical Substances; IECSC - Inventory of Existing Chemical Substances in China; ENCS - Existing and New Chemical Substances (Japan); ISHL - Industrial Safety and Health Law (Japan); PICCS - Philippines Inventory of Chemicals and Chemical Substances; NZIoC - New Zealand Inventory of Chemicals; TCSI - Taiwan Chemical Substance Inventory; CMR - Carcinogen, Mutagen or Reproductive Toxicant; GLP - Good Laboratory Practice

Further information

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific

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material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.

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